FREQUENCY-AWARE HIERARCHICAL ATTENTION AND ARTIFACT-CONSISTENT ORDINAL LEARNING FOR DIABETIC RETINOPATHY GRADING

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ABSTRACT

Reliable diabetic retinopathy (DR) screening requires accurate five-class grading—from subtle early lesions to proliferative disease—and robustness to real-world imaging variability; yet high-frequency lesion cues are weakened by early downsampling and spatial-only pipelines, predictions deteriorate under blur, glare, vignetting, device and population shifts, and standard losses neglect severe class imbalance and the ordinal structure of DR grades. We introduce MFHAN-AR, a frequency-aware hierarchical attention model coupled with robustness- and ordinality-oriented training. The method decomposes each fundus image into low- and high-frequency bands via an adaptive frequency decomposition module, encodes both streams with multi-stage windowed self-attention, and modulates information exchange through frequency cross-gating. A similarity-gated cross-stage refinement aligns semantics across resolutions, and a lightweight fusion head aggregates multi-scale descriptors for final grading. Training combines paired-view artifact consistency, which enforces prediction invariance across views with blur, glare, and vignetting, with a class-balanced focal–ordinal objective that addresses imbalance and penalizes large ordinal errors. Experimental results show that our method achieves competitive performance compared with commonly used CNN baselines, outperforming architectures such as VGG16 and Inception-V3 in several metrics while maintaining comparable overall accuracy and AUC. Ablations indicate that explicit spectral exposure preserves microaneurysms and vascular boundaries, cross-stage refinement improves cross-scale coherence, and the combined loss yields better ordinal calibration than cross-entropy.

1 INTRODUCTION

Deep computer vision systems are increasingly deployed in healthcare to support population-scale screening and triage. Diabetic retinopathy (DR) is a leading cause of preventable blindness, and color fundus photography enables non-invasive detection in primary care settings. Clinical screening relies on reliable five-class grading that ranges from normal to proliferative disease, where decisions depend on both minute, high-frequency cues (e.g., microaneurysms and small dot hemorrhages) and broader context that captures vascular remodeling, ischemic regions, and neovascularization. Real-world images frequently exhibit glare, vignetting, and motion blur; prior treatments such as panretinal photocoagulation introduce laser scars that resemble active lesions. Our goal is to develop a robust, accurate, and deployment-ready five-class grader that preserves fine-scale evidence, integrates global structure, and remains stable across common artifacts, devices, and populations.

Recent advances in convolutional and Transformer-based models have improved long-range reasoning, multi-scale feature aggregation, and semantic understanding for medical images Dosovitskiy et al. (2021); Liu et al. (2021); Shamshad et al. (2023). Hybrid designs that combine convolutional locality with self-attention have shown promise in capturing complementary cues Chen et al. (2021); Zhang et al. (2021); Peng et al. (2021). However, two challenges persist. First, early spatial downsampling and spatial-only pipelines can attenuate high-frequency lesion signatures and sharpened vascular boundaries that are critical for early DR detection. Second, robustness remains fragile: glare, vignetting, and blur can distort appearance; device and population shifts alter color and noise statistics; and treatment-related confounders can mimic active disease, leading to spurious detections

and calibration drift Gulshan et al. (2016); Ting et al. (2017); Abràmoff et al. (2018); Beede et al. (2020). Compounding these issues, five-class DR grading is both highly imbalanced and ordinal. Standard objectives often underweight rare advanced classes and treat all misclassifications equally, ignoring that large-step errors are clinically costlier than near misses Lin et al. (2017); Cui et al. (2019); Cao et al. (2020). These factors limit the reliability and generalization of existing graders in real clinics.

Addressing these gaps is important for both accuracy and trust. Early lesions are subtle; missing or suppressing their signals delays referral and treatment. Conversely, overreacting to artifacts or laser scars undermines specificity and strains care pathways. Ordinal consistency matters for longitudinal care, where small changes should not trigger large jumps in predicted severity. Screening programs span devices, acquisition protocols, and populations; graders that generalize across these axes reduce the need for site-specific tuning and simplify integration into clinical workflows. While related concerns such as calibration and selective abstention are complementary, enabling robust, fine-grained, and order-aware grading is a necessary step toward dependable deployment.

We propose MFHAN-AR, a frequency-aware hierarchical attention network for five-class DR grading. The core idea is to make complementary frequency content explicit and to fuse it across scales so that high-frequency cues are preserved without sacrificing global context. A hierarchical attention backbone aggregates multi-scale evidence while frequency-aware branches expose and emphasize lesion-relevant details that spatial-only features may suppress. To promote stability under common acquisition artifacts and device shifts, we adopt a robustness-oriented training scheme that enforces consistent predictions under artifact-perturbed paired views, encouraging invariance to glare, vignetting, and blur without relying on artifact-specific detectors. Finally, we couple class balancing with an ordinal-aware objective so that learning reflects both rarity and grade ordering, aligning the loss with clinical costs. Together, these design choices target the intertwined challenges of fine-cue preservation, cross-scale semantic coherence, and real-world robustness.

Our main contributions are as follows.

- We introduce MFHAN-AR, a frequency-aware hierarchical attention framework that preserves and fuses complementary low- and high-frequency information for five-class DR grading.
- We propose a cross-scale fusion strategy that aligns semantics across resolutions while retaining lesion-level details critical for early disease.
- We develop a robustness-oriented training scheme that combines artifact-aware consistency with a class-balanced, ordinal-aware objective to address imbalance and label order.
- We provide a comprehensive evaluation with ablations and training-dynamics analyses that clarify the roles of frequency exposure, cross-scale fusion, and robustness training in improving accuracy and stability.

2 Related Work

2.1 Hierarchical Transformers and CNN–Transformer Hybrids for Fundus Analysis

Vision Transformers (ViT) introduced token-based global self-attention as a strong alternative to convolutional backbones for image recognition and are increasingly used in medical imaging Dosovitskiy et al. (2021); Shamshad et al. (2023). Hierarchical designs handle high-resolution inputs and multi-scale structure by progressively reducing token density while maintaining long-range context. Swin Transformer is a representative example, using shifted-window attention and a feature pyramid to obtain near-linear complexity and effective local and global modeling Liu et al. (2021). To reduce attention cost and improve data efficiency in medical settings, decomposed variants such as the Medical Transformer (MedT) apply axial factorization and pair a shallow global branch with deeper local branches Valanarasu et al. (2021). CNN–Transformer hybrids combine complementary inductive biases: sequential couplings such as TransUNet pass CNN features to a Transformer to add global context Chen et al. (2021), whereas parallel dual streams such as TransFuse and Conformer iteratively fuse CNN detail with Transformer context through modules that enable bidirectional exchange Zhang et al. (2021); Peng et al. (2021). Lightweight attention, such as Squeeze-and-Excitation (SE)

and the Convolutional Block Attention Module (CBAM), is often inserted to highlight lesion cues with minimal overhead Hu et al. (2018); Woo et al. (2018).

Relative to this line of work, our MFHAN follows Swin-like hierarchical spatial modeling and a hybrid design, but differs from prior spatial-only backbones by introducing a learnable band-pass decomposition and multi-frequency attention. This enables stagewise fusion of complementary spectral cues together with CNN/Transformer feature exchange.

2.2 Attention, Fusion, and Robustness in DR Pipelines

DR pipelines have long used plug-in attention, such as SE and CBAM, to highlight lesions within convolutional stages, and encoder-level fusion in CNN–Transformer hybrids, such as TransFuse and Conformer, to combine local detail with global context Hu et al. (2018); Woo et al. (2018); Zhang et al. (2021); Peng et al. (2021). These advances improve accuracy but focus mainly on spatial feature interactions. Work on responsible AI in ophthalmology stressed reliability beyond internal accuracy, and large-scale validations and field deployments highlighted the importance of image quality triage, safe referral policies, and handling clinic-specific artifacts and confounders Gulshan et al. (2016); Ting et al. (2017); Abràmoff et al. (2018); Beede et al. (2020). Related studies showed that probability calibration and abstention improve safety Guo et al. (2017); Geifman & El-Yaniv (2019), and that class imbalance and ordinal misclassification costs in five-class DR can be addressed with focal and class-balanced losses and ordinal formulations such as CORAL Lin et al. (2017); Cui et al. (2019); Cao et al. (2020). Adversarial analyses further showed vulnerability to small perturbations, motivating robustness-focused training and evaluation Finlayson et al. (2019).

Within this context, our contribution combines attention and fusion with deployment-oriented training. We incorporate artifact-aware consistency (e.g., blur, glare, vignetting) and couple class-balanced focal and ordinal objectives to improve minority-class recall and reduce large-step errors under realistic screening conditions.

2.3 Spectral/Transform-Domain and Frequency-Aware Representations

Classical fundus analysis used transform-domain methods, such as multi-scale Gabor and wavelet filters, to sharpen vessel edges and detect microaneurysms while suppressing illumination variations, which improved lesion–background contrast Soares et al. (2006); Walter et al. (2007). Recent vision models incorporate spectral operations: FcaNet injects DCT bases into channel attention to modulate low-, mid-, and high-band energy Qiu et al. (2021), FNet mixes tokens with global Fourier transforms at linear cost Lee-Thorp et al. (2021), and Global Filter Networks learn content-adaptive frequency filters Wu et al. (2021). In contrast, popular spatial token mixers and attention modules (e.g., ViT, MLP-Mixer) provide strong global and spatial mixing but do not expose explicit band-pass controls Dosovitskiy et al. (2021); Tolstikhin et al. (2021). These observations suggest that explicit spectral mechanisms can enhance subtle high-frequency retinal cues that are easily reduced by downsampling or cluttered backgrounds.

Our MFHAN builds on this line of work by coupling a learnable band-pass front end with hierarchical windowed Transformer blocks and by introducing multi-frequency attention for dynamic, stagewise fusion of low-, mid-, and high-band cues, thereby preserving micro-lesions and vessel-edge signals while complementing spatial reasoning.

3 Methodology

To address attenuation of high-frequency lesion cues by early downsampling, sensitivity to common fundus artifacts, and the ordinal imbalance in five-class DR grading, we introduce MFHAN-AR, a frequency-aware hierarchical attention model with training tailored for robustness and ordinal constraints. The backbone first separates low- and high-frequency content to preserve subtle microaneurysms and exudates, then integrates local and regional context with windowed self-attention while adaptively modulating frequency channels. Cross-stage refinement encourages semantic consistency across scales, and a compact multi-stage descriptor supports classification. Optimization adds paired-view artifact consistency to stabilize predictions under degradations and a class-balanced focal-ordinal loss to reflect class imbalance and ordinal costs. Given an input image

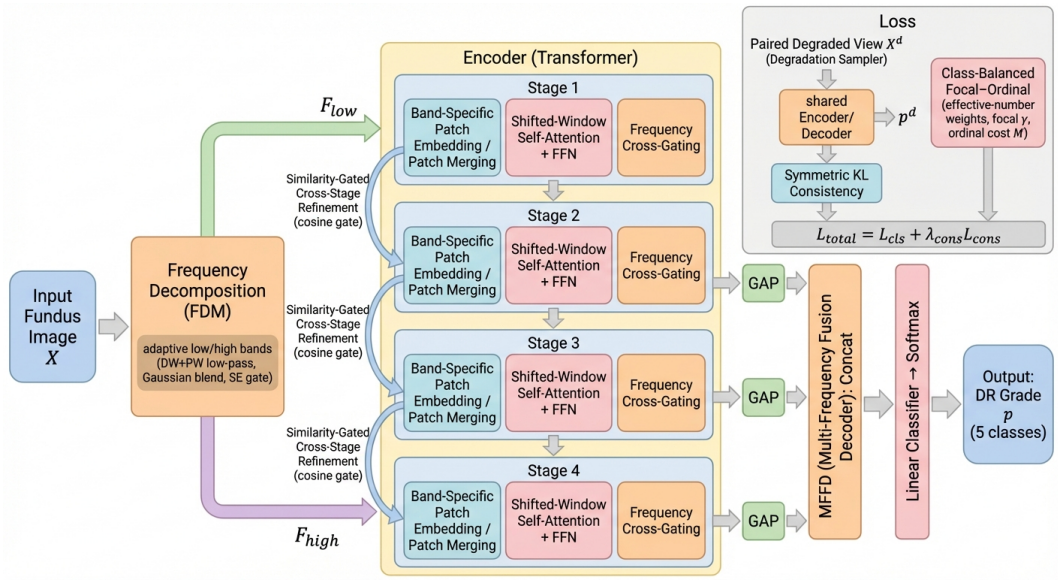

Figure 1: Architecture of MFHAN-AR for five-class diabetic retinopathy grading. The model performs frequency-aware decomposition into low and high bands, applies hierarchical windowed self-attention with frequency cross-gating across four stages, refines features across stages with similarity gating, and fuses multi-stage features for classification. Training uses paired-view artifact robustness and a class-balanced focal-ordinal objective.

$X \in \mathbb{R}^{B\times 3\times H\times W}$, the network outputs logits $Z \in \mathbb{R}^{B\times 5}$ and probabilities $p = \mathrm{softmax}(Z)$ for DR grading.

3.1 Frequency-Aware Decomposition

Clinical intuition suggests exposing spectral bands before early subsampling: early lesions are largely high-frequency and risk being suppressed, whereas broader anatomical context appears in low-frequency content. The Frequency Decomposition Module (FDM) therefore separates low- and high-frequency components with adaptive per-channel gating so that subsequent attention can emphasize lesion-relevant details while retaining structural context. A shallow stem first produces

$$S = \phi\big(\mathrm{BN}(W_s * X)\big), \quad S \in \mathbb{R}^{B\times C_0\times H\times W}, \tag{1}$$

where W_s is a convolution, BN is batch normalization, and ϕ is a pointwise nonlinearity. A learnable low-pass branch is realized with a depthwise-separable convolution,

$$L = W_{l,p} * (W_{l,d} * S), \qquad H = S - \Big[(1-\eta)\,(G_3 * S) + \eta\,(G_5 * S)\Big], \tag{2}$$

where $W_{l,d}$ and $W_{l,p}$ are depthwise and pointwise filters, G_k denotes a depthwise Gaussian blur with kernel size k, and $\eta \in [0,1]$ is a learnable blend of blur scales. A squeeze-excitation gate is derived from the stem via global average pooling (GAP),

$$g = \sigma\big(\mathrm{MLP}(\mathrm{GAP}(S))\big), \quad g \in (0,1)^{B\times C_0\times 1\times 1}, \tag{3}$$

and used to produce frequency-conditioned outputs

$$F_{\text{low}} = g \odot L, \qquad F_{\text{high}} = (1-g)\odot H, \qquad F_{\text{low}}, F_{\text{high}} \in \mathbb{R}^{B\times C_0\times H\times W}. \tag{4}$$

These band-specific features feed the hierarchical encoder without altering spatial fidelity, preserving lesion-scale information for downstream attention.

3.2 Hierarchical Multi-Frequency Attention

The Hierarchical Multi-Frequency Attention Module (HMFAM) uses four stages that integrate local-to-regional context with windowed self-attention while adaptively modulating frequency channels.

Windowed and shifted-window operations constrain attention to local windows with periodic shifts, giving near-linear scaling with image size and maintaining tractability at fundus resolutions. At stage $l \in \{1, \ldots, 4\}$, we maintain a consistent tensor layout (B, C_l, H_l, W_l).

3.2.1 Band-Specific Patch Embedding

To preserve band identity and control aliasing, Stage-1 applies separate patch embeddings to the FDM outputs. Subsequent patch merging reduces spatial resolution while widening channels, expanding receptive fields and enabling multi-scale context aggregation:

$$T^1_{\text{low}} = \text{PE}_{\text{low}}(F_{\text{low}}), \qquad T^1_{\text{high}} = \text{PE}_{\text{high}}(F_{\text{high}}), \tag{5}$$

where PE. are strided convolutions. For $l \geq 1$, a patch-merging operator PM^l reduces spatial resolution and increases channels,

$$T^{l+1}_{\text{low}} = \text{PM}^l\big(T^l_{\text{low}}\big), \qquad T^{l+1}_{\text{high}} = \text{PM}^l\big(T^l_{\text{high}}\big). \tag{6}$$

3.2.2 Windowed Attention with Residual MLP

Within each stage, band tokens are concatenated along channels and processed by shifted-window self-attention followed by a feed-forward network (FFN). This design captures localized lesion patterns and their spatial relationships with efficient attention while preserving stability through residual connections:

$$\begin{aligned} T^l_{\text{cat}} &= \text{Concat}_C\big(T^l_{\text{low}},\, T^l_{\text{high}}\big) \in \mathbb{R}^{B\times 2C_l\times H_l\times W_l}, \\ \hat{U}^l &= \text{Attn}^l\big(T^l_{\text{cat}}\big) + T^l_{\text{cat}}, \qquad U^l = \text{FFN}^l(\hat{U}^l) + \hat{U}^l. \end{aligned} \tag{7}$$

Here Attn^l denotes shifted windowed self-attention with relative position encoding, and FFN^l is a two-layer MLP with activation and normalization.

3.2.3 Frequency Cross-Gating

Artifact-related spectral shifts (e.g., glare or blur) can change the relative utility of low versus high frequency channels. We reweight the low- and high-band channel halves of U^l using band-specific global descriptors, allowing the network to emphasize lesion-bearing bands while suppressing artifact-dominated content. Let $U^l = [U^l_{\text{low}}, U^l_{\text{high}}]$ with $U^l_{\text{low}}, U^l_{\text{high}} \in \mathbb{R}^{B\times C_l\times H_l\times W_l}$. Band summaries are obtained by

$$s^l_{\text{low}} = \text{GAP}\big(T^l_{\text{low}}\big), \qquad s^l_{\text{high}} = \text{GAP}\big(T^l_{\text{high}}\big) \in \mathbb{R}^{B\times C_l\times 1\times 1}, \tag{8}$$

and projected into gates,

$$\begin{aligned} \big[a^l_{\text{low}},\, a^l_{\text{high}}\big] &= \sigma\Big(W^l_g\, \text{Concat}_C\big(s^l_{\text{low}}, s^l_{\text{high}}\big) + b^l_g\Big), \\ V^l &= \text{Concat}_C\big(a^l_{\text{low}} \odot U^l_{\text{low}},\, a^l_{\text{high}} \odot U^l_{\text{high}}\big), \end{aligned} \tag{9}$$

where $a^l_{\text{low}}, a^l_{\text{high}} \in (0,1)^{B\times C_l\times 1\times 1}$. The gated output $V^l \in \mathbb{R}^{B\times 2C_l\times H_l\times W_l}$ is forwarded to the next stage and to cross-stage refinement.

3.3 Similarity-Gated Cross-Stage Refinement

To align semantically consistent structure and suppress redundancy across scales, which is important for linking fine lesions to broader disease patterns, the Cross-Stage Feature Refinement Module (CSFRM) injects a gated residual from V^l into V^{l+1}. A 1×1 projection aligns channels and resolution:

$$R^{l\to l+1} = \Pi^l\big(V^l\big) \in \mathbb{R}^{B\times 2C_{l+1}\times H_{l+1}\times W_{l+1}}. \tag{10}$$

Let $v_p = \text{LN}\big(V^{l+1}_{[:,\cdot,p]}\big)$ and $r_p = \text{LN}\big(R^{l\to l+1}_{[:,\cdot,p]}\big)$ be channel vectors at spatial location p. A location-wise cosine gate is

$$\gamma^l(p) = \sigma\Big(\tau\, \frac{\langle r_p,\, v_p\rangle}{\|r_p\|_2\, \|v_p\|_2 + \epsilon}\Big), \quad \gamma^l \in (0,1)^{B\times 1\times H_{l+1}\times W_{l+1}}, \tag{11}$$

where $\tau > 0$ is a learnable temperature and $\epsilon > 0$ ensures numerical stability. The refined feature is given by

$$\tilde{V}^{l+1} = V^{l+1} + \gamma^l \odot R^{l\to l+1}, \qquad \tilde{V}^{l+1} \in \mathbb{R}^{B\times 2C_{l+1}\times H_{l+1}\times W_{l+1}}. \tag{12}$$

This similarity-gated residual encourages cross-scale consistency while limiting the propagation of misaligned details.

3.4 Multi-Stage Fusion for Grading

The Multi-Frequency Fusion Decoder (MFFD) aggregates refined representations from deeper stages to form a compact descriptor for classification. Global average pooling is applied to stages $l \in \{2, 3, 4\}$,

$$y = \mathrm{Concat}(\mathrm{GAP}(\tilde{V}^2),\, \mathrm{GAP}(\tilde{V}^3),\, \mathrm{GAP}(\tilde{V}^4)) \in \mathbb{R}^{B\times D}, \tag{13}$$

followed by a lightweight linear head producing logits and probabilities,

$$Z = W_c y + b_c \in \mathbb{R}^{B\times 5}, \qquad p = \mathrm{softmax}(Z). \tag{14}$$

This pooling-based fusion yields stable global descriptors suited to ordinal grading.

3.5 Optimization with Robustness and Ordinality

We combine an artifact-consistency regularizer with a class-balanced focal-ordinal classification objective to improve sensitivity to minority classes and penalize errors by grade distance. This pairing reflects clinical priorities: stable predictions in the presence of acquisition artifacts and error costs that increase with grade distance.

3.5.1 Paired-View Artifact Consistency

To enforce prediction invariance under realistic degradations common in retinal imaging (e.g., blur, glare, vignetting; see representative works on robustness), we construct a degraded view X^d by sampling from a distribution of photometric and spatial corruptions and perform a shared-weight forward pass to obtain Z^d and $p^d = \mathrm{softmax}(Z^d)$. In practice, the corruption families include photometric shifts (brightness/contrast, gamma, color casts), illumination artifacts (vignetting, lens glare/flare), noise and compression (Gaussian/Poisson noise, JPEG artifacts), and spatial degradations (defocus and motion blur, mild rotations and cropping), reflecting common fundus acquisition variability. The symmetric consistency loss averages bidirectional Kullback–Leibler divergences:

$$\mathcal{L}_{\mathrm{cons}} = \frac{1}{B}\sum_{i=1}^{B}\Big[\mathrm{KL}\big(p_i \,\|\, p_i^d\big) + \mathrm{KL}\big(p_i^d \,\|\, p_i\big)\Big]. \tag{15}$$

3.5.2 Class-Balanced Focal–Ordinal Loss

To address long-tailed class distributions and encode the ordinal structure of DR grades, we weight examples by effective number and combine focal cross-entropy with an ordinal penalty. Let n_c denote the training count of class $c \in \{1, \ldots, 5\}$. Using effective number weighting , we define

$$w_c = \frac{1-\beta}{1-\beta^{n_c}}, \qquad \beta \in [0, 1), \tag{16}$$

and use focal cross-entropy with optional label smoothing targets $q_{i,c}$:

$$\mathcal{L}_{\mathrm{focal}} = -\frac{1}{B}\sum_{i=1}^{B}\sum_{c=1}^{5} w_c\, q_{i,c}\,\big(1-p_{i,c}\big)^{\gamma_f}\,\log\big(p_{i,c}+\delta\big), \tag{17}$$

where $\gamma_f \geq 0$ and $\delta > 0$ ensure stability. To encode ordinal structure , we penalize the expected quadratic distance between the true grade y_i and predictions using a fixed cost matrix $M \in \mathbb{R}^{5\times 5}$ with

$$M_{ab} = \Big(\frac{a-b}{4}\Big)^2, \qquad \mathcal{L}_{\mathrm{ord}} = \frac{1}{B}\sum_{i=1}^{B}\sum_{c=1}^{5} M_{y_i,c}\, p_{i,c}. \tag{18}$$

The combined classification loss is

$$\mathcal{L}_{\mathrm{cls}} = \alpha_{\mathrm{cls}}\,\mathcal{L}_{\mathrm{focal}} + (1-\alpha_{\mathrm{cls}})\,\mathcal{L}_{\mathrm{ord}}, \qquad \alpha_{\mathrm{cls}} \in [0, 1]. \tag{19}$$

3.5.3 FINAL OBJECTIVE

The network is trained end-to-end with

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{cls}} + \lambda_{\text{cons}} \mathcal{L}_{\text{cons}}, \tag{20}$$

where $\lambda_{\text{cons}} \geq 0$ balances discriminability and robustness. Hyperparameters $\eta, \tau, \gamma_f, \beta, \alpha_{\text{cls}}, \lambda_{\text{cons}}$ are selected on a validation set. We next evaluate whether exposing and modulating frequency bands, enforcing cross-scale coherence, and training for artifact invariance improve ordinal grading under standardized settings and targeted ablations.

4 EXPERIMENTS

4.1 EXPERIMENTAL SETTINGS

4.1.1 DATASETS AND EVALUATION METRICS

We evaluate MFHAN-AR on the APTOS 2019 dataset for five-class diabetic retinopathy (DR) grading, with labels 0–4 corresponding to No DR, Mild, Moderate, Severe, and PDR. To respect the ordinal structure and maintain robustness, we use an 80/20 stratified train/validation split and report validation results only in this study. Performance is evaluated on the validation set using multiple metrics, including accuracy (Acc), quadratic weighted kappa (QWK), precision, recall, F1-score, and AUC. Among these metrics, QWK is particularly important for ordinal classification tasks as it penalizes predictions according to their distance from the ground truth labels.

4.1.2 BASELINES

To evaluate the effectiveness of the proposed method, we compare it against several widely used convolutional neural network architectures that are commonly adopted for image classification tasks. Specifically, we consider ResNet50, VGG16, Inception-V3, and MobileNet-V3 as baselines. These architectures represent different design paradigms, including deep residual networks, classical convolutional networks, multi-scale convolutional structures, and lightweight mobile-oriented models. All baseline models are trained under the same experimental protocol to ensure fair comparison. The models are optimized using the same training set, data preprocessing pipeline, and evaluation procedure. Our method (denoted as Ours) follows the same training configuration and evaluation protocol as the baseline models.

4.1.3 IMPLEMENTATION DETAILS

We train MFHAN-AR with AdamW (initial learning rate 3e-4, weight decay 0.05), mixed precision, and gradient clipping. We run 100 epochs with a fixed learning rate (no warmup or cosine schedule) and a batch size of 8, using a WeightedRandomSampler for class balancing. The primary loss is a Class-Balanced Focal–Ordinal objective (CBFOCriterion; beta=0.999, gamma_f=2.0, eps=0.05, alpha=0.7), with a CE-only variant for ablation. When ART is enabled, a symmetric KL consistency term (lambda_cons=0.5) is applied to paired original/degraded views produced by stochastic Gaussian blur, motion blur, vignetting, glare, and laser scar simulation. The model includes an FDM for low/high band decomposition with channel-wise SE gating and optional frequency dropout (p_freqdrop=0.2), HMFAM stages with anti-aliased BlurPool downsampling and Swin-like local attention (with optional frequency cross-gating), CSFRM via cosine-similarity gates, and an MFFD head for classification.

4.2 MAIN PERFORMANCE COMPARISON

Table 1 presents the quantitative comparison between our method and the baseline architectures. Among the baseline models, ResNet50 achieves the strongest performance, obtaining an accuracy of 58.2% and a QWK score of 0.651. This result confirms the effectiveness of deep residual learning for this task. Our method achieves an accuracy of 52.7% and a QWK score of 0.528, which is comparable to the performance of VGG16 and Inception-V3. In particular, our model slightly outperforms VGG16 in accuracy (52.7% vs. 52.5%) while producing similar AUC scores (0.772 vs. 0.771). Compared with Inception-V3, our method provides improvements across several metrics,

Table 1: Quantitative comparison with baseline CNN architectures on the validation set.

Method	Acc ↑	QWK ↑	Precision ↑	Recall ↑	F1 ↑	AUC ↑
ResNet50	0.582	0.651	0.584	0.581	0.578	0.808
VGG16	0.525	0.570	0.524	0.526	0.517	0.771
Inception-V3	0.519	0.536	0.509	0.519	0.509	0.761
MobileNet-V3	0.445	0.362	0.431	0.446	0.432	0.717
Ours	0.527	0.528	0.520	0.527	0.519	0.772

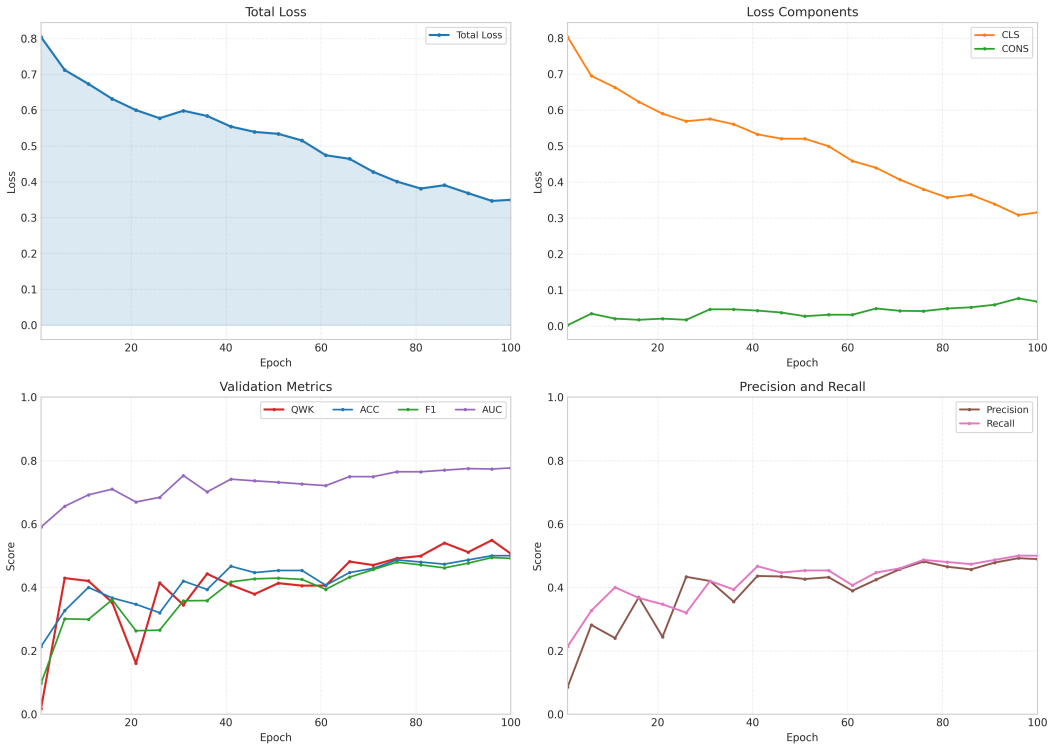

Figure 2: Learning curves over 100 epochs.

including accuracy (52.7% vs. 51.9%), F1-score (0.519 vs. 0.509), and AUC (0.772 vs. 0.761). Lightweight models such as MobileNet-V3 perform substantially worse, achieving only 44.5% accuracy and 0.362 QWK, indicating that model capacity plays a critical role in this task. Overall, while ResNet50 remains the strongest baseline, the proposed method demonstrates competitive performance compared with commonly used CNN architectures such as VGG16 and Inception-V3, suggesting that our approach provides a reasonable trade-off between model complexity and predictive performance.

4.3 Training Dynamics and Convergence

We track training loss and validation signals across 100 epochs to characterize optimization behavior and convergence in Figure 2. Over training, the model shows a rapid improvement in the early stage: QWK increases from 0.02 at epoch 1 to around 0.40 by epoch 10, accompanied by consistent gains in ACC and AUC. As training proceeds, the optimization becomes more stable and the training loss gradually decreases from 0.80 to about 0.55, while validation performance continues to improve with moderate fluctuations. In the later stage, the model exhibits clear convergence behavior. The training loss further decreases to around 0.38–0.40, and the validation metrics stabilize, with QWK exceeding 0.50, ACC reaching about 0.50, and AUC approaching 0.77. Overall, the curves indicate a stable optimization process and consistent performance improvement throughout training.

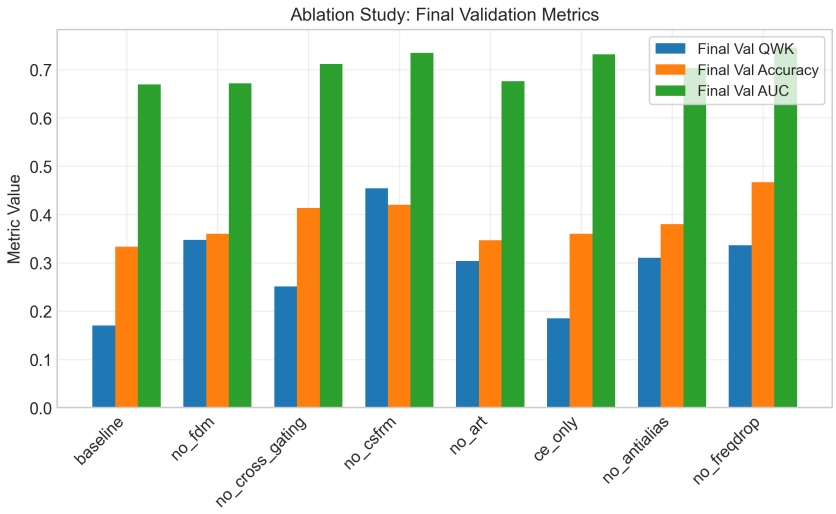

Figure 3: Ablation and optimization traces over 30 epochs. The full model shows loss reduction (0.865→0.575) and a best validation accuracy of 45.3% at epoch 29.

Table 2: Ablation study of different components in the proposed method. The full model integrates all modules.

Method	Acc ↑	QWK ↑	Prec ↑	Rec ↑	F1 ↑	AUC ↑
CE Only	0.500	0.500	0.498	0.500	0.486	0.736
w/o Anti-Alias	0.453	0.437	0.446	0.453	0.449	0.709
w/o ART	0.507	0.386	0.488	0.507	0.484	0.761
w/o Cross-Gating	0.493	0.498	0.480	0.493	0.482	0.745
w/o CSFRM	0.527	**0.533**	0.511	0.527	0.508	**0.778**
w/o FDM	**0.540**	0.509	**0.541**	**0.540**	**0.531**	0.775
w/o FreqDrop	0.500	0.524	0.477	0.500	0.478	0.751
Ours (Full Model)	0.527	0.528	0.520	0.527	0.519	0.772

4.4 Ablation Studies

To understand the contribution of each component in the proposed framework, we conduct a series of ablation experiments on the validation set. Starting from the full model, we progressively remove individual modules or training strategies while keeping all other settings unchanged. The evaluated components include the proposed Cross-Scale Frequency Refinement Module (CSFRM), Feature Distillation Module (FDM), cross-gating interaction, Adaptive Representation Transformation (ART), frequency-domain regularization (FreqDrop), anti-aliasing filtering, and the combined training objective. Table 2 reports the ablation results. The full model achieves 52.67% accuracy and 0.528 QWK, providing balanced performance across metrics. Using cross-entropy only degrades performance (Acc 50.0%, QWK 0.500), indicating the benefit of the proposed training objective. Removing anti-aliasing causes the largest drop (45.33% Acc, 0.437 QWK), highlighting the importance of suppressing aliasing artifacts. Removing FreqDrop also reduces performance, suggesting that frequency-domain perturbation improves generalization. For architectural components, removing cross-gating lowers accuracy to 49.33%, showing its role in feature interaction. Removing ART significantly reduces QWK (0.386), suggesting weaker ordinal consistency. Removing FDM slightly increases accuracy (54.0%) but reduces QWK and F1, indicating less consistent predictions. Removing CSFRM results in comparable performance but does not improve overall metric balance. Overall, each component contributes to the final performance, and the full model provides the most stable results.

5 CONCLUSION

We address five-class diabetic retinopathy grading, where subtle micro-lesions, artifacts, and an imbalanced ordinal label space demand models that preserve detail and remain stable. We introduce MFHAN-AR, a frequency-aware hierarchical attention network that exposes and fuses low- and high-frequency cues across scales while enforcing artifact-consistent views. Training with a class-balanced focal-ordinal loss further biases learning toward clinically meaningful, small-step errors. we investigated the effectiveness of our proposed model through a comprehensive comparison with standard CNN baselines. The results demonstrate that our approach achieves competitive performance against established architectures, suggesting its potential as a practical alternative for this classification task.

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